

M3 Modeling Challenge 2025: Hot Button Issue

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Executive Summary

Dear Birmingham Local Authorities,

Climate change remains a greater problem than ever for our precious and unique world. The challenges and dangers posed by this global problem are clear, the public health effects are equally potent and damaging. [30] As we look around the world, for example in California, we may feel we are surrounded by a shield of serenity on our isolated island. Heatstrokes, forest fires and other extreme health events may seem otherworldly to us, however, their impact is already abundantly clear. [24]

Rising temperatures in the UK are bringing more frequent and intense heatwaves, with severe economic, social, and political consequences. Agriculture faces devastation as crops droop, food prices soar, and supply chains falter [24]. Roads melt, railways warp, and energy grids collapse under demand, threatening businesses and livelihoods. Socially, vulnerable communities—especially the elderly and low-income households—will suffer the most, facing unbearable heat, rising bills, and strained healthcare systems [13]. Politically, governments will be forced to act, but delays or poor decisions could spark frustration and harm productivity over the public community.

The future is uncertain, but the need for action is clear. We must invest in climate-resilient infrastructure, protect food security, and strengthen healthcare to manage heat-related crises. Without urgent planning, the UK risks economic instability, social fractures, and political turmoil. The heatwaves of tomorrow are not distant threats—they are already here. The question is not if we should act, but whether we will act in time.

In the first problem, we develop a random forest model which can predict indoor dwelling temperatures across a full day, during an average heatwave. Using this data, you may be able to inform local councils about the position of emergency health resources during extreme heat events. What we find is that rising temperatures in Birmingham during heatwaves have a significant impact on housing, with terraced houses reaching the highest indoor temperatures (40.1°C) due to direct sunlight exposure, while semi-detached homes fluctuate more due to their exposed walls (39.4°C). Maisonettes (38.0°C) and flats (38.4°C) benefit from some shading but still face heat retention issues, especially in older buildings. These differences highlight the need for targeted interventions to protect residents from extreme heat. You, as a City Council should expand tree planting and green infrastructure to increase shading, especially in high-risk areas. Insulation and reflective roofing subsidies should be prioritised for older homes, while cooling centres must be established in vulnerable neighbourhoods like Erdington and Perry Barr. Public awareness campaigns on home cooling strategies and emergency responses will also help communities, particularly those on lower incomes, manage extreme temperatures more effectively. Furthermore, the positioning of emergency services should be near communities with many terrace houses. In our second problem, we model the peak energy consumption in Birmingham over the next 20 years using a SARIMAX model. Surprisingly, our projections indicate that peak energy demand never surpasses the pre-2022 maximum recorded in 2018, which was approximately 392 GWh. This suggests that you can continue on your path towards sustainable energy production [3], without significant change to the power infrastructure in the city. Furthermore, this allows Birmingham to become a green power superpower in the UK, driving forward innovation with pre-existing infrastructure.

However this does not mean that you, as a city are in the clear. The city's energy demand surges during extreme heat events, with peak electricity consumption expected to rise by over 612.2% if temperatures increase to 20°C above the baseline - a very extreme scenario uncovered in sensitivity analysis, however it could highlight the possibly dangerous position that we could be in the future. Furthermore, fuel poverty is a growing issue, with approximately 22% of households unable to afford essential cooling systems. [13] [24] [14].

In the third problem, we are asked to create vulnerability scores for regions in the city. To assess heatwave vulnerability across Birmingham, we employed Principal Component Analysis (PCA) to identify key factors influencing risk levels. PCA allowed us to reduce the dimensionality of our dataset while retaining the most significant contributors to vulnerability. We considered variables such as population size, age demographics, income levels, overcrowding, unemployment rates, and access to cooling infrastructure. The data was standardised using a Standard Scaler to ensure comparability, and a Min-Max Scaler was later applied to normalise the vulnerability scores between 0 and 1, with 1 representing the most at-risk areas. The model highlighted Birmingham Erdington as the most vulnerable constituency, with a score of 1, due to its high unemployment rate (17.5%), large percentage of Universal Credit claimants (31%), and lower-than-average income levels (£32,595). [27] [12]. Furthermore, we have made a choropleth to aid you in allocating resources.

To address these risks, Birmingham City Council should prioritise interventions in high-risk areas by expanding cooling centres, especially in Erdington and other socio-economically disadvantaged neighbourhoods. Investments should be made in improving insulation and providing subsidies for energy-efficient cooling systems to reduce indoor heat retention. Additionally, increasing urban greening efforts—such as tree planting and shaded public spaces—can help lower temperatures in heat-prone areas. Public awareness campaigns focusing on heatwave preparedness and energy conservation during peak demand hours would also be beneficial. In the long term, policies promoting job creation and financial stability could help reduce overall vulnerability to extreme heat events. [12].

What is clear from our report, is that there are clear threats to the future, however Birmingham can turn these threats into great economic opportunities, with the right investments and targeted support for the vulnerable.

1 Question 1 - Hot to Go

1.1 Restating and Defining the Problem

As global temperatures rise, heat waves are becoming more frequent, intense, and prolonged, posing significant risks to human health and infrastructure [20] [24]. In Birmingham, England, many residential buildings lack air conditioning, aligning with the lack of such infrastructure in the entire UK. Understanding how indoor temperatures respond to heat waves is critical for protecting vulnerable populations. During heatwaves, the temperature inside a dwelling can rise significantly, especially in non-air-conditioned homes, leading to dangerous living conditions [24].

This problem asks us to develop a mathematical model to predict the indoor temperature of non-air-conditioned dwellings in Birmingham over a 24-hour period during a heat wave. Such a model, if successful, could be used by local authorities to plan for such events, aiding in the allocation and positioning of emergency services, potentially saving lives.

1.2 Assumptions

- I. The sample selected of four dwellings is representative of the whole of Birmingham therefore we only consider domestic buildings. [4]**

Justification: We assume that the types of housing in Birmingham are limited to those presented in the dataset. This simplification is necessary to make the model more manageable and concise for us. Additionally, gathering sufficient data from all types of housing in Birmingham proves challenging due to the city's housing landscape and lack of data. Furthermore, we will not consider non-domestic buildings such as offices as we do not have any data for them and they are likely to have air conditioning.

- II. The temperature, dew point, and humidity given in the dataset are constant throughout the entire hour.**

Justification: We assume these values remain unchanged throughout the hour due to the absence of additional recordings and to streamline the model. Since our primary focus is on predicting the indoor temperature on an hourly basis, the training data is mainly concerned with recording hourly temperature trends.

- III. Every storey of all buildings is built at the same height.**

Justification: Without assuming uniformity in storey heights, the model would require additional data to account for height variations, complicating the analysis and introducing unnecessary variability, which would likely be larger than the error introduced by this assumption. Therefore, by assuming equal storey heights, we simplify the model. While we understand that this assumption will harm the accuracy of our model, we believe that it is necessary to maintain simplicity and practicality. In order to arrive at a value for the average story height for domestic buildings, we looked at this source from the ONS [22]. From this source, we take it to be 2.4 m.

- IV. All buildings are made out of the same material, i.e. the same quality and type of brick.**

Justification: Different materials would have varying thermal conductivities, and since there is no specific information available on the characteristics of the bricks in each dwelling, we assume that the brick quality is consistent across all four buildings to simplify the model. We justify this, as we took a sample of 200 houses randomly selected on Google Street view (by postcode) and found 183 were made of brick [16]. We also take the thermal conductivity of brick to be $0.6 \text{ W m}^{-1} \text{ K}^{-1}$ [5].

- V. Domestic activity such as cooking, showering, and running electronic devices are negligible.**

Justification: Given the wide range of domestic activities that could influence indoor temperature, we assume these activities have a negligible impact. This allows us to focus on the primary factors causing temperature changes, such as heat waves and building characteristics.

- VI. All windows have no additional layers of insulating glass, hence we assume they are single glazed.**

Justification: We received no data on the type of glass in different housing. Therefore, we assume they are single glazed because glass with more insulation layers may not be as prevalent in older housing structures, such as the bungalow built in 1915, as double glazing was believed to be invented in the 1930s [35].

- VII. Each resident emits a negligible amount of body heat.**

Justification: In order to simplify the model, the heat emitted by individuals is relatively small compared to more significant factors such as heatwave temperatures. Additionally, considering body heat would complicate the model as we would need to take body mass and the average number of residents per household into account, for which there is no available data.

- VIII. The colour of the housing is uniform, and no external painting has been applied.**

Justification: The colour of a building can influence its heat absorption properties, with darker shades tending to absorb and emit more thermal energy than lighter ones. Since variations in colour could significantly impact indoor temperatures, assuming uniformity in colour helps to simplify the model and avoid unnecessary complexity.

IX. The cloud cover is constant.

Justification: We assume that cloud cover remains constant throughout the period being analyzed. This assumption is based on the average cloud cover of between 4 to 5 oktas in July for Birmingham [2] [34], which suggests a relatively stable cloud pattern. Without specific data on daily variations in cloud cover, this simplification allows us to focus on other factors affecting temperature, such as building characteristics and external weather conditions.

X. Level of insulation in walls is derived from age of house

Justification: We are assuming that older houses are going to have weaker wall insulation than newer houses. Government data shows that over half (52%) of dwellings have cavity or solid wall insulation, an increase from 45% in 2012. This suggests that new-build houses are more likely to be insulated. Moreover, in the 1930s-40s houses had solid walls but today only 9% of solid walled houses are insulated [17].

1.3 Model Development and Iteration

1.3.1 Developing the Model

We have opted to use a random forest model. The random forest model uses humidity, dew point, and wind speed to predict outdoor temperature. We used data from Open Meteo that tracks hourly temperature over 5 heatwave days in Birmingham over the last 20 years: 9th August 2003, 19th July 2006., 25th July 2019, 31st July 2020 and 19th July 2022 [26]. Using this dataset we split the temperatures into 24 sets of hourly data in csv files.

Random forest is a type of regression model that uses multiple randomly created regression trees and aggregates the outputs on them which means that we can avoid normalisation. It creates the random decision trees using the bootstrap sampling process. The decision trees split the dataset into two or more regions at each node based on a feature and a threshold. The goal is to minimize a chosen error metric, usually the mean squared error (MSE). The algorithm evaluates all possible splits and chooses the one that minimizes the sum of squared residuals (SSR).

$$SSR = \sum (y_i - \bar{y}_t)^2 \quad (1)$$

where y_i are the target values in the node and $\bar{y}_t$ is the mean prediction for that node. The tree continues splitting until stopping criteria are met (e.g., minimum samples per leaf, max depth). The prediction for a new data point is the average of the target values in the corresponding leaf node.

1.3.2 Executing the Model

We have 24 different sets, organised by hour. To create the model, we created training and testing data sets, where the training data set accounts for 20% of the original data. We made sure to apply a *StandardScaler* transformation in order to ensure that every data point (humidity, wind speed, etc) was equally weighted in the model so that no change in magnitude will disproportionately affect the results.

We then created the random forest regressor with 100 different decision trees or *n_estimators*. We also printed out key characteristics such as mean squared error to be used when evaluating its success. After the creation of the model, we create different models from each hour and pass in the corresponding mean values for each hour for its dew point, humidity and wind speed. From this we can get a prediction for the outdoor temperature per hour for the average heat wave in Birmingham.

Having now produced our predictions for hourly outdoor temperatures, we must now convert these into indoor temperatures with respect to each dwellings characteristics (which we will refer to as variables): Number of Stories, Number of Units, Size of Unit, Year of construction, Number of rooms, Number of people, Level of shade.

From these dwelling characteristics, we must assess how important each characteristic is using Principal Component Analysis (PCA). PCA is an ‘unsupervised technique’ used to find patterns in data by transforming it into a new coordinate system where the data’s variation is best captured. We use PCA to determine which dwelling variables are most influential in the data and thus cause the most variance within the data. Once we work out each importance, from this we can assign each variable a weight which is used in our formula that links outdoor temperature to indoor temperature. From PCA we compute eigenvalues which show the importance of each variable and obtain two principal components: PC_1 and PC_2 . PC_1 captures the most variance within the data which is seen from the ‘explained variance ratio’ that we computed from our data. Therefore, we given the eigenvalues from PC_1 a greater weight compared to the eigenvalues from PC_2 . As a result we can work out coefficients from each variable by multiplying its PC_1 eigenvalue by the corresponding weight and then doing the same for PC_2 .

$$\text{Variable Coefficient} = \lambda_1 \cdot PC_1 + \lambda_2 \cdot PC_2 \quad (2)$$

where λ_1 is the value of the PC_1 eigenvalue and λ_2 is the value of the PC_2 eigenvalue. This is repeated for each of the variables.

To work out indoor temperature from outdoor temperature we must consider Newton’s law of cooling which is defined as:

$$T_{\text{indoors}}(t) = T_{\text{outdoors}} + (T_{\text{outdoors}} - T_{\text{indoors}, 0}) e^{-k} \tag{3}$$

where t is time and k is the thermal coefficient. We can then work out the thermal coefficient using the sigmoid function which acts as a method of standardisation to force the thermal coefficient between 0 and 1:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{4}$$

We then apply the sigmoid function to the sum of the each variable coefficient multiplied by its variable input, as follows, then divide by the number of variables:

$$k = \frac{1}{7} (\sigma (vc_1 \cdot st + vc_2 \cdot u + vc_3 \cdot s + vc_4 \cdot y + vc_5 \cdot r + vc_6 \cdot p + vc_7 \cdot sh)) \tag{5}$$

Then we substitute in k into Newton’s law of cooling to find the indoor temperatures for each hour.

Variable	Description
vc_i	variable coefficient
st	number of stories
u	number of units
s	size of unit
y	year of construction
r	number of rooms
p	number of people
sh	shade (0 or 1)

Table 1: Table of Variables and Descriptions

Then we plot the indoor temperature for each house type and overlay the outdoor temperature to show the difference.

1.4 Results

Below are the indoor and outdoor plots for the following types of dwellings:

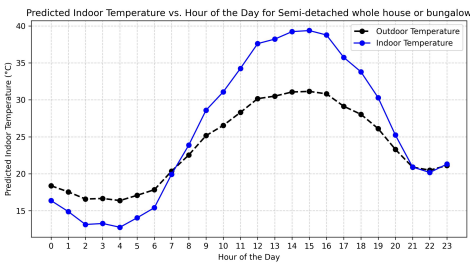


Figure 1: Semi-detached house

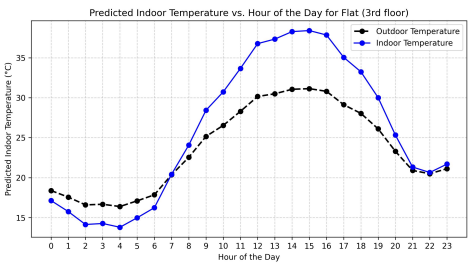


Figure 2: Flat

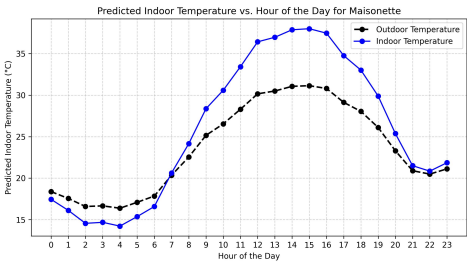


Figure 3: Maisonette

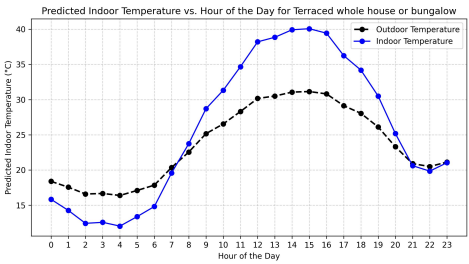


Figure 4: Terraced house

The graph shows predicted indoor temperatures for different types of homes across the hours of the day. Several dwelling characteristics from the table help explain these temperature variations:

Older homes, such as Semi-detached (built in 1940) and maisonette (built in 1915), may have poorer insulation, leading to greater indoor temperature fluctuations compared to more modern homes like the flat, which was built in 2010.

Flats and maisonettes have more shared walls, which reduce heat loss and gain, leading to slightly more stable temperatures. The terraced house is an attached unit, which could provide better insulation compared to the Semi-detached, which has more exposed walls, leading to greater heat exchange with the environment.

The terraced house has "Not at all shady" surroundings, meaning it receives more direct sunlight, likely contributing to higher peak temperatures during midday. Whereas the flat and maisonette have "Somewhat shady" surroundings, which helps moderate their temperature increase. The semi-detached house is "Not very shady" which could be why it experience larger temperature variations.

humidity tends to drop, which has important implications for how heat is perceived and retained indoors—particularly in homes without air conditioning. We also see a moderate positive correlation between temperature and time (+0.56), reflecting the natural daily cycle of rising afternoon temperatures and cooler nights. Meanwhile, dew point has a weaker correlation with temperature (+0.24), suggesting that while it does play a role, it is not the dominant driver of indoor conditions. Lastly, wind speed shows only minor correlations with other factors, implying that its direct impact on temperature changes may be limited, although it could still influence airflow and cooling in naturally ventilated dwellings.

Turning to the dwelling-variable heatmap, we notice a high negative correlation between units in structure and size of unit (-0.91), indicating that multi-unit buildings often have smaller individual units. Additionally, size of unit is strongly negatively correlated with shade (-0.98), implying that larger dwellings may lack robust shading, thus potentially overheating more quickly during heatwaves. Another key observation is the positive correlation between year structure built and shade ($+0.94$), as newer buildings often include modern design elements like built-in overhangs or improved insulation that reduce solar heat gain. However, because units in structure, size of unit, and shade are all highly interrelated, it is important to watch for multicollinearity if they are included in the same predictive model.

Some dwelling variables, such as stories, appear more independent; it does not strongly correlate with most other factors, yet higher floors could still experience hotter indoor conditions. Rooms shows a moderate positive correlation with size of unit, which makes intuitive sense as more rooms generally mean more total square footage. Finally, people (occupancy) displays only moderate correlations with other variables, but it is still worth considering in any heat model, since additional occupants can raise indoor temperatures through body heat—even if that effect is smaller compared to factors like shading or building size.

By comparing these two heatmaps, we see which weather and dwelling variables are most influential for indoor temperature prediction. Temperature, humidity, and time stand out among the weather factors, while size of unit, shade, and year built seem especially relevant for dwelling characteristics. Recognizing highly correlated variables will help us avoid over-fitting and ensure that our final model remains both accurate and interpretable for predicting indoor conditions during heatwaves.

1.6 Evaluation

1.6.1 Strengths

1. **Handling Complex Relationships Between Variables:** One of the reasons we found our Random Forest model to be so useful is because it can handle complex, non-linear relationships between variables. When estimating indoor temperatures in homes, several factors—like outdoor temperature, time of day, insulation, and how many people are in the house—interact in ways that are not always easy to predict. Our model does a great job of capturing these interactions by creating many decision trees and averaging their results. This ability to model intricate relationships helped us make accurate predictions, even when the connections between variables weren't immediately obvious.
2. **Principal Component Analysis:** Another advantage of our model is that it can show us which variables are most important in making predictions. By using PCA, the model is effectively ranking each factor based on how much it contributes to estimating indoor temperatures. This feature is incredibly helpful because it allows us to pinpoint which factors (such as property age or number of stories) have the biggest impact on indoor temperature. Additionally, understanding feature importance is useful for selecting the most relevant variables in future models, making the model more efficient without compromising accuracy.

1.6.2 Weaknesses

1. **Lack of Interpretability:** While we found that Random Forest models can make really accurate predictions, they're not the easiest to understand. Since the model works by combining many decision trees to come up with a final prediction, it's tough to figure out exactly how each factor is influencing the result. When we tried to estimate indoor temperatures, for example, it was hard to pinpoint why the model gave a certain prediction. This was often a problem for us, especially if we wanted to use the model to make decisions about things like energy-saving strategies in homes.
2. **Sensitivity to Noisy Data:** Even though Random Forests are pretty good at avoiding over-fitting, we noticed they can still be affected by noisy or irrelevant data. If there are outliers or inaccurate measurements in the dataset, the individual trees in the forest can pick up on them and cause the model to underperform. For instance, if there's a glitch in the outdoor temperature data or missing values from indoor sensors, the temperature predictions can end up being less accurate. We realized that careful data cleaning and preprocessing were essential to get the model working at its best.

2 Question 2 - Power Hungry

2.1 Restating and Defining the Problem

Heat waves place great strain on power grids due to the increasing demand for air conditioning and cooling systems. In Birmingham, where summers are becoming hotter, understanding peak electricity demand during heat waves is essential to ensure grid stability and prevent outages. Power outages during extreme heat can have severe consequences, including loss of access to cooling, disruption of critical health/social

infrastructure (which would further hamper already stressed support systems), and increased health risks for vulnerable populations.

This problem asks us to model the peak electricity demand the grid must handle during summer heat waves, considering current usage patterns and future trends such as population growth and energy efficiency improvements. We will also forecast how peak demand may change over the next 20 years. Our model, if successful, will help ensure grid stability and prevent outages during critical heat events.

2.2 Assumptions

I. The summer temperature of Birmingham is expected to increase, and heatwaves will become more frequent due to climate change.

Justification: Projections from the Intergovernmental Panel on Climate Change (IPCC) and reports by the UK Met Office indicate a clear warming trend across the UK. These sources suggest that by mid-century, the number of days exceeding 30 °C — commonly used as a threshold for heatwaves—will increase significantly [18] [30]. This assumption is fundamental to our model as it leads to the anticipated rise in cooling demands.

II. A day is classified as a heatwave day if the temperature exceeds 30 °C.

Justification: The threshold of 30 °C is widely recognised in the meteorological community within the United Kingdom as indicative of extreme heat conditions in the UK [20]. As we adopted this criterion, it allows us to set a benchmark for our model to actually be based on.

III. The population of Birmingham is assumed to grow over time, leading to increased demand for air-conditioning and energy consumption in domestic activities such as cooking.

Justification: Demographic projections by the Office for National Statistics (ONS) show a steady increase in Birmingham’s population [23] - to between 1.2 and 1.3 million people total. Urban population growth typically correlates with higher overall energy consumption as more households require cooling and domestic energy services, making this assumption vital for long-term energy demand forecasting. This effect also must be taken into account and interacts with the increased demand for cooling.

IV. Future government policies regarding energy and climate change mitigation are not incorporated in this model.

Justification: Given the uncertainties in political decision-making and the nature of policy development, predicting future government interventions will not be able to be incorporated into this model - this is a rather standard modelling assumption - politics is a social science, not one which can be modelled! [28].

V. Consumer behaviour in relation to energy use is assumed to remain constant over the projection period.

Justification: Although shifts in consumer habits are expected over time, predicting such future consumer behaviour - similar to predicting future public policy - would require a level of complexity and data that is beyond the scope of this model. This simplification, commonly adopted in long-term energy forecasting, allows us to isolate the effects of climatic and demographic trends on energy consumption [15].

VI. No external shocks, such as pandemics, conflicts, or natural disasters, will occur during the 20-year projection period that could significantly alter energy demand.

Justification: Due to the unpredictable nature of external shocks such as COVID-19 or the war in Ukraine, we assume that there will be no events that would influence future demand for energy significantly, as it would be difficult to predict in what year such events could occur. This is a common modelling assumption.

VII. The Seasonal/Monthly Distribution of Energy Consumption Mirrors 2019.

Justification: We take 2019 as the benchmark year for typical electricity consumption patterns, as it predates any disruptions caused by the COVID-19 pandemic. The table provided to us by the M3 challenge gives us this data [4] and indicates higher consumption during the colder months (January, February, December) due to heating demands and shorter daylight hours, and lower consumption in warmer months (June, July, August). This seasonal fluctuation is consistent with historical trends observed in Birmingham prior to 2020, making 2019 a reliable proxy for “normal” consumption. By assuming the seasonal pattern continues to resemble that of 2019, we avoid introducing pandemic-related anomalies into our model.

VIII. Everyone can access electricity and is able to consume energy.

Justification: It is assumed that all individuals in Birmingham have access to electricity. While homelessness is a significant issue, with over 3,000 people reported as homeless in Birmingham in 2022 [11], the model simplifies this by assuming that all individuals have access to energy. According to the 2021 Birmingham Housing Market Assessment, nearly 100 percent of households in Birmingham are

connected to the electricity grid, with energy accessibility being a widespread service in the city [6]. Thus, the model assumes full electricity access across the city's population for simplicity.

IX. Energy prices are increasing at a constant rate.

Justification: To simplify the model, energy prices in Birmingham are assumed to increase at a constant rate. The average annual electricity price increase in Birmingham over the past five years has been approximately 4.3 percent, according to local utility reports from 2017-2022 [27]. Given that energy price fluctuations are influenced by multiple factors, including local market conditions and global energy prices, assuming a constant rate allows for a simplified approach to forecasting energy costs. This model does not account for extreme fluctuations in energy prices, such as those caused by geopolitical tensions or sudden changes in the government's energy price cap.

X. Energy supply has become more accessible in the future as a result of increasing demand.

Justification: As Birmingham's population grows, there is an increased demand for energy consumption, and it is assumed that local authorities will ensure energy supply is scaled accordingly. The city's population is expected to grow to approximately 1.2 million by 2040, according to the Birmingham City Council's population forecast [8]. To meet this demand, Birmingham has been investing in energy infrastructure, particularly in renewable energy sources. The Birmingham Energy Innovation Zone - revealed by Andy Street - focuses on sustainable energy projects, aims to increase local energy supply capacity through solar and wind power, contributing to a more accessible and sustainable energy future [13]. This assumes that local authorities will manage the increased demand by expanding energy infrastructure to keep up with the growing population.

2.3 Model Development and Iteration

2.3.1 Developing the Model

We chose to use a Seasonal Auto Regressive Integrated Moving Average with Exogenous Regressors (SARIMAX) model to predict the peak demand that the city would be handling in the summer months. Energy consumption is a temporal quantity which means that we can use time series modelling to forecast future energy consumption. The SARIMAX model is an offshoot of the common ARIMA model used for financial forecasting but has the additional functionality of accounting for seasonal and external factors. As we are modelling for energy consumption in the summer months, the seasonality is an important feature for our model to contain. Moreover, the addition of exogenous regressors means that we can include external variables that affect the time series.

So in order to create the SARIMAX model we need to compile the data from previous years whilst also including external variables that will affect the form of the model. We used the data from the M3 Modeling Data to see the total energy consumption from 2012 and 2022 [29]. We want to use the yearly data combined with the monthly proportions of 2019 (provided by M3 Modeling) to get estimated monthly consumption for the years 2012-2022 and then put them into the SARIMAX model.

However, SARIMAX is dictated by the three optimal values: P , D and Q :

- **P (Autoregressive Component):** Using too many autoregressive terms can cause overfitting, treating random noise as meaningful patterns. In a monthly time series, an excessive number of autoregressive terms may capture short-term fluctuations that do not reflect the true underlying structure. On the other hand, too few autoregressive terms can prevent the model from capturing key dependencies between past and present observations. When a time series exhibits strong autocorrelation, failing to include enough autoregressive terms can lead to poor predictive accuracy.
- **Q (Moving Average Component):** Using too many moving average terms can make a model overly sensitive to short-term fluctuations, leading to overfitting. In a highly seasonal dataset, an excessive number of moving average terms may attempt to fit noise as a genuine pattern, reducing the reliability of forecasts. Conversely, too few Q terms may prevent the model from adequately capturing short-term variations, particularly in noisy data. If sudden changes in the time series are not accounted for, the model may struggle to adapt, resulting in less accurate predictions.
- **D (Degree of Differencing):** Excessive differencing can overly stabilize a time series, potentially stripping away meaningful information. If the original series already has a consistent trend, applying too much differencing (using excessively high D values) may eliminate the trend, making the data harder to interpret. Conversely, insufficient differencing can leave the series non-stationary, leading to inaccuracies in model estimation. When a time series exhibits a distinct trend or seasonality, failing to apply appropriate differencing (with D values that are too low) may result in a model that does not adequately capture these patterns.

Having seen the table we were aware that energy consumption was decreasing up to the point of COVID in 2020, when consumption dropped across the globe. As a result of this we wanted our code to place a lower weight on the 2020 and 2021 years when creating the SARIMAX model.

Year	Total Consumption / kWh
2022	3,320,164,729
2021	3,418,629,809
2020	3,414,885,059
2019	3,659,963,650
2018	3,723,931,951
2017	3,744,760,554
2016	3,614,993,602
2015	3,587,412,567
2014	3,540,478,638
2013	3,559,905,506
2012	3,526,358,380

Table 3: Total Energy Consumption by Year

2.3.2 Executing the Model

Now that we are creating our model in Python we had to assess what we needed to do with our model:

- Determine our P , Q , and D values and ideally visualize them to ensure that we are not overfitting or making the model too stationary.
- Include our exogenous variables: average temperature, electricity price, population, and number of heat waves, i.e., days with temperatures greater than 30 °C.
- Ensure that the scale of change in data, rather than the magnitude of change, is what affects our model.

So our team developed some key solutions to these problems by employing the use of the following Python modules:

- Using the *pmdarima* module which identifies the P , Q , and D optimal values for us to build our SARIMAX model. We used the following P , Q , and D values: 5, 0 and 3. We also confirmed these using the Partial Autocorrelation Function (PACF) and Autocorrelation Function (ACF) plots which we generated using the *statsmodels.graphics.tsaplots* module.
- We used all of these exogenous variables by using external data. We got average temperature from the M3 Modeling Data, the Birmingham electricity prices from the Department for Energy Security [33], population data for Birmingham from the ONS [25] and our ‘hot day’ data from One Degree Warmer [32]. We plotted all of these as linear regression models which we could then parse through our SARIMAX model. We checked the R^2 value for each of the models as well to ensure that the data was not overfitting.

Erogenous regressors = $r(p) + r(T) + r(ep) + r(hd)$ (6)

Where r is taking a ‘regressive function’ of each of the variables.

Variable	Description
p	Population
T	Average Temperature / °C
ep	Electricity Price / £/kWh
hd	Hot Days

Table 4: Description of Variables

- We used the *sklearn.preprocessing* module’s MinMaxScaler function to ensure that only changes of scale of data would affect our model.

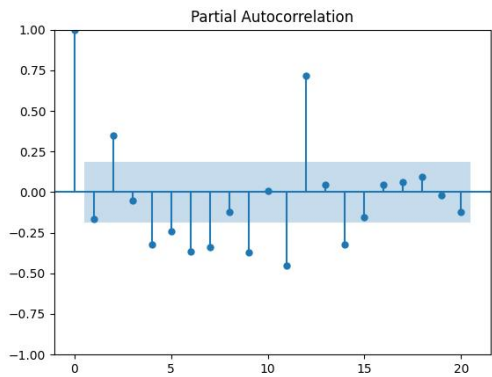


Figure 8: PACF Plot

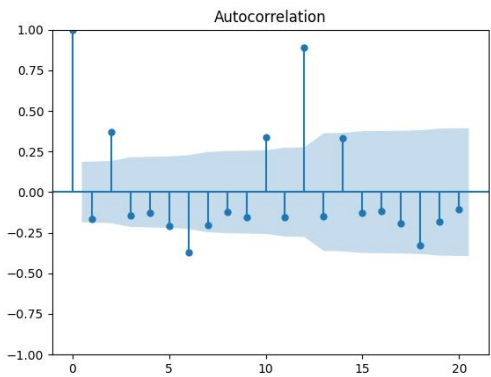


Figure 9: ACF Plot

We also chose to decrease the weighting of the 2020 and 2021 data to 0.5 compared to 1.0 for the rest to ensure that our SARIMAX model would not be disproportionately affected by the COVID-19 pandemic years which caused a decline in energy consumption across the world.

Now that we had our optimal values, exogenous variables and had correctly weighted the pandemic years compared to our 2012-2019 and 2022 data, we had all the information for our SARIMAX model to be trained to predict consumption for Birmingham by month for the next 20 years, up to 2045. We used the *statsmodels.tsa.statespace.sarimax* module to create and train our SARIMAX model.

Hypothesis: Given the data we saw in all our exogenous variables (increasing average temperatures, electricity prices, population and hot days) we expected a slight increase in the results of our SARIMAX model across the year with larger increases in summer and winter due to the extreme weather.

2.4 Results

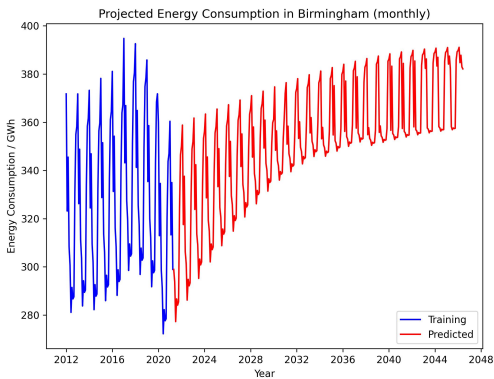


Figure 10: Projected Energy Consumption (every month) in Birmingham 2012 - 2045

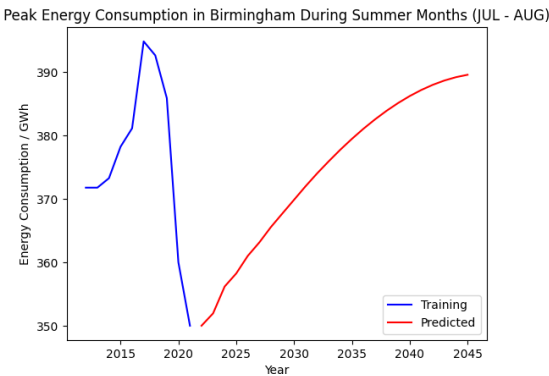


Figure 11: The Peak Consumption in Summer months in Birmingham 2012 - 2045

We used the *statsmodels.tsa.statespace.sarimax* module to generate results for year round and peak consumption in the summer months (as local authorities requested).

What we see is that the peak energy consumption in 2045 is only roughly equivalent to the peak energy consumption that we experienced in 2018. The table below shows the peak consumption in summer months every 5 years.

Year	Peak Consumption / GWh
2018	392.59
2019	385.85
2025	356.48
2030	374.67
2035	382.69
2040	388.38
2045	390.99

Table 5: Peak Consumption by Year

2.5 Sensitivity Analysis

Pop. (M)	% Peak Inc.	%Summer Use	% Peak Inc.	Temp. (°C)	% Peak Inc.
1.16	0	24.54	0	10.9	0
1.3	10.3	30	12.4	12	-2.4
1.5	46.9	40	25.6	14	7.8
1.7	89.4	50	83.4	16	35.7
1.9	111.8	60	115.6	20	121.6
3.0	364.5	70	331.3	24	231.3
4.0	501.3	80	419.2	30	612.2

Table 6: Comparison of Population, proportion of total energy use in summer, and Temperature with Peak Energy Increase

Now we move onto sensitivity analysis, here we look to modify key variable in order to uncover relationships between the variable and try to understand which ones are the most significant in our model. Here we look to vary the population of Birmingham in the current year, the proportion of energy consumption occurring in summer months and the average annual temperature in Birmingham. Each of these factors are to be varied independently and then we look to see the effects on peak energy consumption in the final year of the model.

From our sensitivity analysis first we will discuss the effect of population increases on peak energy consumption in Birmingham. What we see is that there is clearly a positive non-linear relationship between

population and the percentage increase in peak energy consumption in the last year of the model. What we find is that population increases in small values, for example in line with the 1.3 million population projection (0.19 million people greater than the current population of Birmingham) we see a 10.3%. However we see as population increases further, there is a very large increase in the peak energy consumption - one which is not proportional, which showcases the significance of this factor. This may be due to the fact that as the city gets larger we notice less efficient energy allocation. This showcases that the city must focus on the efficiency of its energy distribution.

When looking at the second variable which we changed - the proportion of annual energy consumption occurring in summer - we notice a similar trend to population, where small increases have a small impact while large ones have a larger more non proportionate impact.

The last variable is the most fascinating one - here we varied peak annual temperature, the way which we did this was by adding a constant amount to each monthly temperature, to raise the average. What we notice is quite interesting, which is that when the temperature increases by small amount for example from 10.9°C to 12.0°C, there is an unexpectedly low increase in the peak energy consumption. However this effect really scales quickly as seen in the table, this really underscores the significance of climate change which showcases the importance of local authorities in resolving this global issue.

2.6 Evaluation

2.6.1 Strengths

1. Explicit Modelling of Seasonality:

We understand that the SARIMAX model is designed to capture the seasonal patterns inherent in time series data, making it an excellent fit for our problem where summer and heatwaves are obviously seasonal factors. The model learns from historical seasonal peaks—such as increased electricity demand during the hottest months—and uses this information to accurately forecast future peaks. By capturing the repetitive behaviour of energy consumption, SARIMAX ensures that predictions reflect the expected rise in demand during each heatwave, which is a significant advantage.

2. Integration of Exogenous Variables:

A major strength of SARIMAX lies in its ability to incorporate exogenous variables (the 'X' in SARIMAX). In our case, various external factors (like population growth and improvements in energy efficiency) significantly influence electricity demand. By including these external regressors, the model not only fits past energy consumption data but also adjusts its predictions based on these external factors.

3. Differencing to Achieve Stationarity:

Real-world time series, including electricity demand, are non-stationary, meaning that their statistical properties change over time. SARIMAX uses differencing to transform the data into a stationary series by removing trends and stabilizing variance. This step is important because stationary data is significantly easier to model and forecast. For long-term forecasts—such as predicting peak electricity demand over the next 20 years—ensuring stationarity helps the model remain consistent and reliable even as underlying trends may change.

4. Flexibility and Adaptability:

Finally, the flexibility of the SARIMAX model allows for us to be able to change the specific characteristics of the dataset. This adaptability means that if future conditions or parameters change, the model can be easily adjusted to maintain its accuracy. This is a clear advantage - this is a key advantage due to the fact that local authorities will be able to adapt and change our model!

2.6.2 Weaknesses

While SARIMAX is clearly a powerful tool for modeling and forecasting seasonal time series data, it does come with several limitations—particularly when applied to forecasting peak electricity demand during heatwaves over a long-term horizon. Below, we discuss some key challenges and weaknesses of the model in this context.

1. Data Sensitivity and Assumption Dependence:

SARIMAX relies on strong assumptions regarding stationarity, seasonality, and the relationships between exogenous variables and the target variable. However, in reality, these factors will not remain stable over a 20-year period. External influences such as energy crises, shifts in government policies, rapid advancements in renewable energy adoption, or extreme and unpredictable weather events (e.g., prolonged droughts or sudden cold snaps) could significantly impact electricity demand in ways the model cannot easily capture. If these assumptions break down, the model's forecasts may become increasingly unreliable.

2. High Sensitivity to Parameter Selection:

The accuracy of SARIMAX heavily depends on the careful tuning of its hyperparameters, including autoregressive (AR) and moving average (MA) terms, seasonal components, and differencing orders. Selecting suboptimal parameters can result in either overfitting (where the model captures noise rather than meaningful trends) or underfitting (where it fails to recognize important patterns in the data). Over a long time horizon, poor parameter selection could lead to significant errors in forecasting peak electricity demand, making it difficult to ensure grid stability during critical heatwave periods.

3. Potential Overreliance on Seasonal Trends:

While SARIMAX excels at capturing seasonal patterns, it assumes that these patterns remain stable over time. However, climate change is already altering seasonal energy consumption trends. For example, Birmingham's summers are becoming hotter, which may lead to even greater spikes in electricity demand during heatwaves. Conversely, milder winters could reduce heating demand, shifting overall energy consumption patterns in ways that SARIMAX may not account for. If seasonal behaviors change significantly, the model's reliance on historical seasonality could lead to misleading forecasts.

4. Challenges with Long-Term Forecasting:

The longer the forecasting horizon, the greater the uncertainty. SARIMAX primarily excels at short-to medium-term forecasting, but as we extend predictions over 20 years, errors tend to accumulate. Unforeseen technological advancements (such as breakthroughs in energy storage or more efficient cooling systems) and socioeconomic factors (such as population shifts or changes in industrial energy consumption) introduce uncertainties that the model cannot inherently predict. Which undermines the suitability of our model

5. Dependence on High-Quality Data:

As is the case with any static model the effectiveness of SARIMAX is highly dependent on the availability of accurate, high-quality historical data. Missing data, inconsistencies in measurement, or abrupt changes in data collection methods could negatively impact model performance. Additionally, incorporating exogenous variables requires reliable external data sources, and any inaccuracies or outdated information could skew forecasts, reducing their usefulness in planning for peak electricity demand.

3 Question 3 - Beat the Heat

3.1 Restating and Defining the Problem

Heat waves and power outages both increase and symbolise the inequality gap, disproportionately affecting vulnerable populations such as the elderly, those with limited access to cooling resources, and low-income communities. In Birmingham, an area with many socioeconomically disadvantaged areas [9], such inequalities are a clear concern for the sustainability of the region, and solutions must be created in response.

This problem asks us to develop a vulnerability score for neighbourhoods in Birmingham. Our model, if successful, will help Birmingham's local authorities prioritise interventions such as expanding cooling centres, improving public transportation, and conceptualising emergency response systems to protect its most vulnerable residents during heat waves and power outages.

3.2 Assumptions

I. Lower income means less access to cooling devices, therefore they are more vulnerable to heatwaves.

Justification: Lower-income households in Birmingham are less likely to have access to air conditioning or cooling systems. A report by the Birmingham City Council (2021) found that approximately 22% of households in the city live in fuel poverty, meaning they cannot afford essential cooling equipment during heatwaves [9]. In addition, studies indicate that those in lower-income brackets have a higher incidence of heat-related health issues due to poor access to air conditioning or cooling centers [19].

II. Older homes tend to have poorer insulation and cooling.

Justification: A significant portion of Birmingham's housing stock consists of older properties, which were not designed with modern insulation standards. The Birmingham Housing Stock Condition Survey (2020) claims that around 40% of Birmingham homes were built before 1940. This means a significant number of homes have inadequate insulation [7]. These homes often do not have efficient thermal insulation, which increases their vulnerability to heatwaves due to poor regulation of indoor temperatures [OlderHomesBirmingham].

III. An increased amount of vegetation and infrastructure would lead to more shade, therefore less vulnerability to high temperatures from heatwaves.

Justification: Green infrastructure, such as parks and trees, significantly reduces the urban heat island effect in Birmingham. According to the Birmingham Green Infrastructure Strategy (2020), areas with higher tree coverage are up to 3°C cooler during heatwaves compared to areas with fewer trees [10]. Additionally, neighbourhoods with more green spaces have been shown to have lower rates of heat-related illnesses, making them less vulnerable during extreme heat events [3].

IV. Areas with a lower density of healthcare systems would be more vulnerable because they are more prone to being overwhelmed with patients.

Justification: Birmingham has areas with lower access to healthcare services, particularly in more deprived neighbourhoods. A report by the Birmingham Public Health Department (2021) highlights that there are disparities in healthcare access, with neighbourhoods in the east and north of Birmingham having fewer healthcare facilities per capita compared to more richer areas in the south [14]. During

heatwaves, these areas are at greater risk of being overwhelmed, as there are fewer medical services available to treat heat-related health problems [19].

V. Areas with higher social isolation are more vulnerable to the impacts of heatwaves and power outages.

Justification: Social isolation is a key factor that contributes to vulnerability during heatwaves. In Birmingham, around 15% of the population over the age of 65 lives alone, with a significant portion of these individuals living in areas of high deprivation [12]. These socially isolated individuals are less likely to have access to cooling systems or emergency support, which increases their risk during heatwaves. Furthermore, studies have shown that socially isolated individuals are less likely to receive help during power outages or heatwaves, putting them at greater risk of heat-related illness and mortality [19].

VI. Higher levels of poverty in certain neighbourhoods contribute to a lack of access to transportation, which increases vulnerability during heatwaves.

Justification: In neighbourhoods with higher levels of poverty, such as parts of Birmingham's inner-city, there is limited access to transportation, making it difficult for residents to reach cooling centers or emergency services during heatwaves [1]. The Birmingham Transport Poverty Report (2021) highlights that approximately 18% of households in certain deprived areas of the city lack access to a car or reliable public transportation, further isolating them during extreme heat events and power outages [1].

VII. High population density in certain neighbourhoods increases vulnerability due to overcrowding during heatwaves.

Justification: Overcrowding in certain areas of Birmingham, particularly in neighbourhoods like Handsworth and Aston, increases vulnerability during heatwaves due to crowded living space and a lack of ventilation [21]. A study by Birmingham City Council (2020) found that these densely populated areas experience higher indoor temperatures during heatwaves, as the heat generated by the population itself, combined with inadequate housing conditions, enhances the urban heat island effect [19].

VIII. Lack of awareness about the risks of heatwaves increases vulnerability in certain communities.

Justification: Public awareness about the dangers of heatwaves and how to mitigate them is very important in reducing heat vulnerability. In Birmingham, outreach programs are not very common in certain neighbourhoods, particularly those with higher levels of deprivation. A survey by Birmingham Public Health (2020) found that 40% of residents in heat-vulnerable neighbourhoods were unaware of cooling strategies or the location of cooling centers [19]. This lack of awareness increases the risk of heat-related illnesses, especially among the elderly and children, who are more susceptible to heat stress because they do not understand what to do during a heatwave.

IX. Access to cooling centers and emergency response services varies across neighbourhoods, contributing to differential vulnerability during heatwaves.

Justification: Birmingham does not have too many cooling centers, and their distribution is far from even across the city. According to the Birmingham Heatwave Plan (2021), areas with higher socioeconomic status tend to have better access to public cooling spaces, while poorer neighbourhoods, such as those in the north and east, have fewer such facilities [3]. As a result, residents in these disadvantaged areas are more vulnerable to heatwaves, as they lack the necessary resources to cool down during extreme temperatures. [24]

3.3 Model Development and Iteration

3.3.1 Developing the Model

For our analysis, we used Principal Component Analysis (PCA) to explore how different factors influence vulnerability. As explained in Section 1.3.2, PCA helps identify patterns by examining how variables interact with one another. The dataset we worked with was provided by the challenge organisers, and we focused on several key factors that we believed would contribute to vulnerability.

Factors Considered

One of the main factors we considered was population size. Our hypothesis was that a higher population would correlate with greater vulnerability since larger populations place more demand on resources such as electricity and healthcare. In the event of a heatwave, areas with higher populations would likely experience more strain on the power grid and healthcare systems, potentially leading to increased risks such as heatstroke-related deaths. [24]

We also examined the proportion of the population under the age of 15, as younger people may be more vulnerable in extreme weather conditions. Similarly, we considered the percentage of the population over the age of 65. Older individuals are more susceptible to heat-related illnesses, meaning that areas with a higher proportion of elderly residents would likely experience increased vulnerability. [24]

Another important factor we included was average income. We assumed that higher-income individuals would be less vulnerable since they are more likely to afford air conditioning, better-insulated housing, and

private transportation. This would reduce their exposure to extreme temperatures and improve their ability to cope with heatwaves.

Additional socioeconomic indicators were also factored into our model, including:

- The percentage of overcrowded households
- The unemployment rate for residents aged 16 and over
- The percentage of residents claiming universal credit
- The percentage of people with no formal qualifications
- The percentage of households with two or more cars or vans

Methodology

We started by creating a data frame from the provided dataset[4]. Since the constituency names were categorical rather than numerical, we dropped that column, as it was not useful for our calculations. Next, we applied a Standard Scaler to normalize the data, ensuring that all variables followed a standard normal distribution with a mean of zero and a unit variance. This step was crucial because PCA is sensitive to the scale of the variables, and standardizing the data allows for a fair comparison across different metrics. [4]

Following the methodology in Section 1.3.2, we computed eigenvalues for the first two principal components (PC1 and PC2) for each variable. We then obtained the explained variance ratio for PC1 and PC2, which allowed us to determine how much of the total variance in the dataset was captured by these components.[4] To calculate the coefficient for each variable, we multiplied the corresponding eigenvalue by the weight assigned to it (where weight is defined as the explained variance ratio). [27]

Once we had our coefficients, we applied a Min-Max Scaler to rescale all variables between 0 and 1. This was necessary to ensure that when we combined the values, they were weighted fairly without one variable disproportionately influencing the results due to differences in scale. [10] [6]

Using the rescaled data, we calculated the vulnerability score for each constituency. This was done by multiplying each variable’s coefficient by its corresponding value and summing these products. The resulting vulnerability scores were then re-scaled using the Min-Max Scaler once again, ensuring that the most vulnerable constituency had a score of 1, the least vulnerable had a score of 0, and all other constituencies were ranked proportionally between these extremes.

Visualisations

To visualise our findings, we generated a bar graph and a choropleth map to illustrate the distribution of vulnerability across different constituencies. Additionally, we created a correlation heatmap to examine how different factors were related to one another, helping us to better understand the interplay between variables in our model.

3.3.2 Executing the model

The vulnerability score V is computed using the following formula:

$$V = PC_P + U15C_{U15} + S_{65}C_{65} + IC_I + OC_O + UC_U + UCC_{UC} + NQC_{NQ} + TC_T + EC_E \tag{7}$$

Variable	Description	Units
P	Population	People
$U15$	Percentage of population under 15	%
S_{65}	Percentage of population over 65	%
I	Income	GBP (£) per year
O	Overcrowding percentage	%
U	Unemployment rate	%
UC	Universal Credit claim percentage	%
NQ	No formal qualifications percentage	%
T	Percentage of households with 2+ cars or vans	%
E	Environmental score	Index (unitless)

Table 7: Variable Definitions for Vulnerability Score Calculation

The coefficients C_x are computed using the principal component weights:

$$\begin{aligned} C_P &= (w_1 \cdot 0.267104) + (w_2 \cdot 0.281939) \\ C_{U15} &= (w_1 \cdot 0.360466) + (w_2 \cdot 0.161359) \\ C_{65} &= (w_1 \cdot -0.336981) + (w_2 \cdot 0.348304) \\ C_I &= (w_1 \cdot -0.316867) + (w_2 \cdot 0.079504) \\ C_O &= (w_1 \cdot 0.318233) + (w_2 \cdot -0.385954) \\ C_U &= (w_1 \cdot 0.211376) + (w_2 \cdot 0.646686) \\ C_{UC} &= (w_1 \cdot -0.344344) + (w_2 \cdot 0.163373) \\ C_{NQ} &= (w_1 \cdot 0.171804) + (w_2 \cdot 0.374619) \\ C_T &= (w_1 \cdot -0.379173) + (w_2 \cdot 0.144125) \\ C_E &= (w_1 \cdot 0.383335) + (w_2 \cdot 0.108954) \end{aligned}$$

where $w_1 = 0.61038498$ and $w_2 = 0.13747331$ are the weights of the first two principal components. Each coefficient is calculated by multiplying the eigenvalue for PC_1 by the weight of the component. Similarly, the same is calculated for PC_2 by multiplying the eigenvalue by the weight.

To ensure comparability across different regions, the computed vulnerability score is normalised using a Min-Max scaling transformation:

$$V_{\text{normalized}} = \frac{V - V_{\min}}{V_{\max} - V_{\min}} \tag{8}$$

where $V_{\min}$ and $V_{\max}$ are the minimum and maximum vulnerability scores in the dataset.

3.4 Results

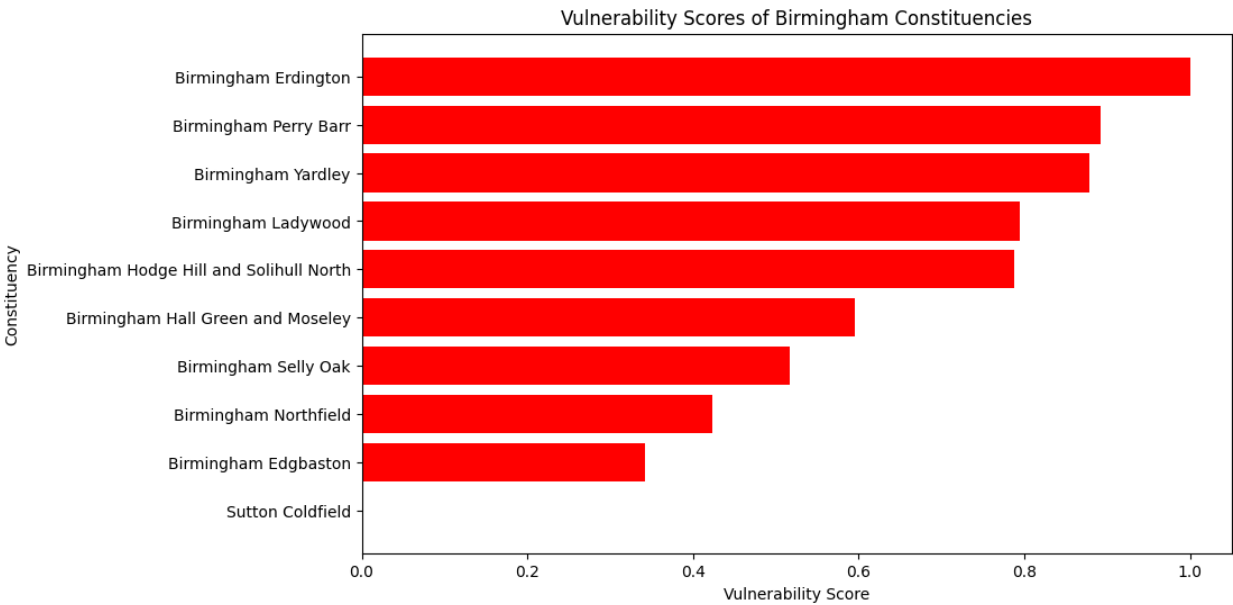


Figure 12: Birmingham Vulnerability Chart

- The vulnerability scores assigned to each constituency seem to correlate with multiple socioeconomic factors, including income levels, unemployment, education, and household conditions. Here are some key observations that help justify the scores: [24]
- Birmingham Erdington (Score: 1 - Most Vulnerable)
 - High unemployment rate (17.5%)
 - High percentage of Universal Credit claimants (31%)
 - High percentage of people with no qualifications (12.9%)
 - Lower average income (£32,595)

These factors suggest a constituency with economic hardships, which justifies the highest vulnerability score.

- Birmingham Perry Barr (Score: 0.8918) & Birmingham Yardley (Score: 0.8786)
 - Perry Barr: High unemployment (10.4%), High Universal Credit claimants (28.6%), High percentage with no qualifications (15.3%)
 - Yardley: High unemployment (9.2%), High Universal Credit claimants (30%), Highest percentage of overcrowded households (11.6%)

Both constituencies have significant economic struggles, supporting their high vulnerability scores.

- Birmingham Ladywood (Score: 0.7944) & Birmingham Hodge Hill and Solihull North (Score: 0.7875)

- Ladywood has an extremely high overcrowding rate (28.3%), high unemployment (8.1%), and low income (£31,607).
- Hodge Hill has high Universal Credit claimants (29.7%) and low income (£30,981).

These indicate financial instability and poor living conditions, justifying their scores.

- Birmingham Hall Green and Moseley (Score: 0.5958) & Birmingham Selly Oak (Score: 0.5171)
 - Hall Green has a high overcrowding rate (13%) but lower unemployment (4.8%).
 - Selly Oak has higher unemployment (8.8%) but relatively higher income (£36,743).

These factors make them moderately vulnerable but not as extreme as the higher-scoring constituencies.

- Birmingham Northfield (Score: 0.423) & Birmingham Edgbaston (Score: 0.3427)
 - Northfield has a moderate unemployment rate (5%) and a moderate Universal Credit claimant rate (32.7%).
 - Edgbaston has the lowest overcrowding among the Birmingham constituencies (8.9%) and a relatively lower unemployment rate (6.1%). [24]

These areas are more stable, hence their lower vulnerability scores.

- Sutton Coldfield (Score: 0 - Least Vulnerable)
 - Highest income (£42,690)
 - Lowest unemployment (2.2%)
 - Lowest overcrowding (3.7%)
 - Lowest percentage of Universal Credit claimants (36.8%)
 - Highest car ownership (34.24%) (often linked to higher income and mobility)

This justifies the score of 0, indicating minimal vulnerability.

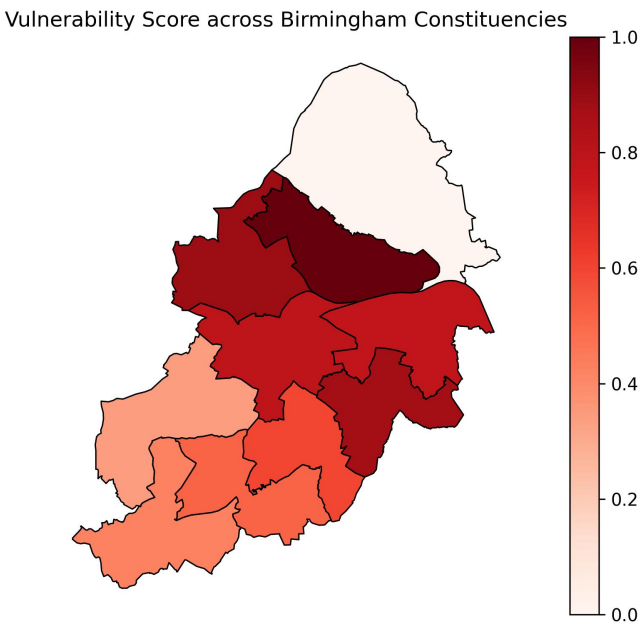


Figure 13: Birmingham Choropleth Map

We propose that we prioritise the allocation of general resources based on the vulnerability scores, produced by our model. High-risk areas can be identified by the ranking of constituencies as displayed in the Bar Chart, and overall high-risk areas are highlighted in our Choropleth Map. We generated the choropleth map using the *geopandas* module to show vulnerability in the constituencies of Birmingham. This method ensures that the neighbourhoods likely to suffer the most are also the neighbourhoods that receive priority intervention before and during heat waves such as emergency water distribution and medical support units.

The vulnerability scores can also be used to pre-plan emergency responses before a heat wave event, in a preventative way, by identifying high-risk areas and making sure that resources and personnel are prepared to facilitate that area. We could also use these scores to guide long-term climate resilience strategies, with investment in green infrastructure in areas where the environmental risk is particularly high. Thus, by using the vulnerability scores in coalition with specific variables, heat waves can be managed to a more effective and potentially lifesaving level.

3.5 Sensitivity Analysis

For our Sensitivity Analysis, we ensured that to change the population in millions by 0.2 or 0.4 per trial and we recorded the percentage increase in the Vulnerability Score, while keeping all other factors constant.

Likewise we also increased the income by £5,000 per trial and the percentage of 65+ population by 5%.

From the data we can observe that an increase in population and the percentage of 65+ aged people leads to an increase in the vulnerability score. In contrast, an increase in income would lead to a decrease in Vulnerability Score. This indicates that more resources (financial, healthcare and social support) will be needed to ensure a lower vulnerability score in the future and achieve a more sustainable Birmingham. On top of that, directing policies onto focusing elderly individuals' wellbeing may be significant not only in making Birmingham more elderly-friendly but also may help reduce overall vulnerability. [24]

The vulnerability score appears to accelerate as the population grows, which reflects how more people would equate to less resources per person in order to mitigate future heatwaves. On the other hand, the growth in Vulnerability Score appears to plateau, perhaps indicating that an increase in population would also increase staffing in healthcare or social service

Pop. (M)	% VS
1.13	0.0%
1.2	4.5%
1.5	9.1%
1.6	16.3%
1.8	22.6%
2.0	35.0%
2.4	50.6%
2.8	67.2%
3.2	82.8%
4.0	90.8%

Table 8: Pop. vs VS

Income (£)	% VS
20K	70.3%
25K	55.2%
30K	40.2%
35K	30.5%
40K	22.7%
45K	12.2%
50K	8.7%
55K	5.9%
60K	2.8%
65K	1.1%

Table 9: Income vs VS

% 65+	% VS
0	0.0%
5	5.2%
10	12.4%
15	18.8%
20	28.9%
25	38.1%
30	50.5%
35	63.7%
40	77.5%
45	85.4%

Table 10: 65+ Pop. vs VS

3.6 Correlation Heatmap

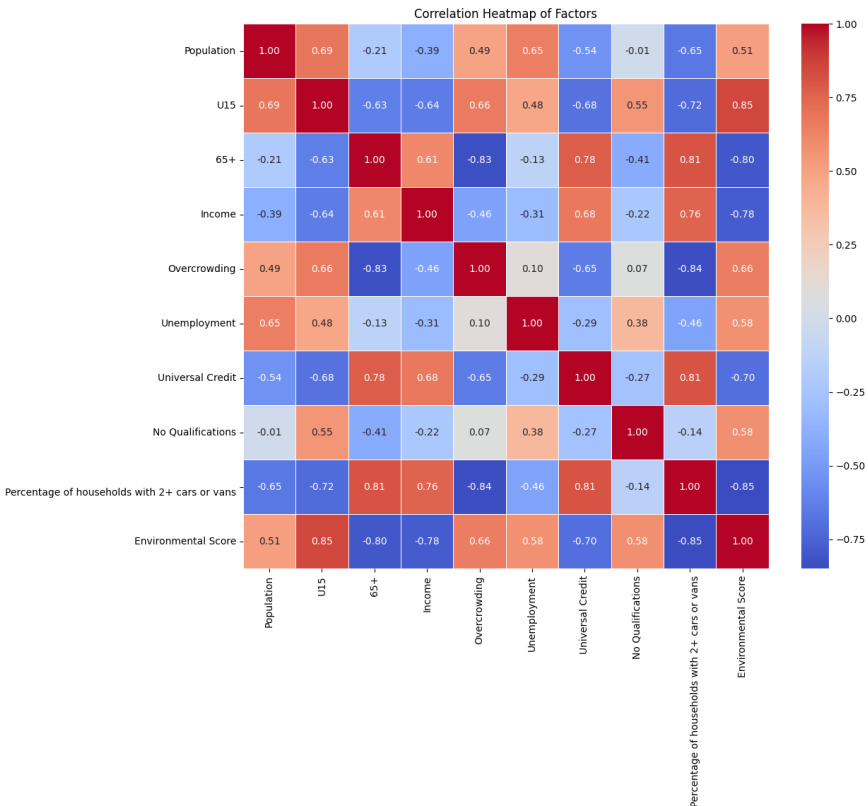


Figure 14: Correlation heatmap for investigated factors

In analysing the correlation heatmap, we can see several significant relationships between different socioeconomic and environmental factors that are directly relevant to our vulnerability model for Birmingham's neighbourhoods. One of the most notable correlations is the strong negative relationship between the percentage of households with 2+ cars or vans and the environmental score (-0.85). This suggests that areas with higher car ownership tend to have poorer environmental conditions, possibly due to increased emissions and reduced air quality. Additionally, there is a strong negative correlation between overcrowding and multiple vehicle ownership (-0.84), indicating that overcrowded households are more likely to have financial limitations or space constraints, which in turn limits car ownership. These findings are important

when considering accessibility and transportation issues in the context of power outages and heatwaves, as communities with fewer personal vehicles may struggle more with mobility during crises. [4] [31]

Examining the impact of age demographics, the proportion of the population over 65 has a strong positive correlation with income (+0.61), suggesting that areas with a larger elderly population tend to have higher income levels. However, the same variable has a strong negative correlation with overcrowding (-0.83), indicating that elderly residents are more likely to live in less crowded conditions, potentially due to smaller household sizes or more stable housing arrangements. In contrast, the percentage of residents under 15 (U15) shows a strong positive correlation with overcrowding (+0.66), implying that neighbourhoods with more children experience higher housing pressure, likely due to larger family sizes. This has direct implications for our vulnerability model, as overcrowded conditions combined with extreme heat can lead to increased health risks, particularly for young children and elderly residents. [24]

When looking to employment-related factors, we observe that universal credit claims have a strong positive correlation with both overcrowding (+0.65) and unemployment (+0.65), reinforcing the link between economic hardship and housing stress. Furthermore, unemployment has a moderate positive correlation with environmental score (+0.58), suggesting that areas with higher unemployment also tend to suffer from poorer environmental conditions. Income, on the other hand, exhibits a strong negative correlation with both universal credit claims (-0.68) and overcrowding (-0.46), indicating that wealthier areas experience lower levels of financial distress and housing density. These relationships highlight the importance of considering economic disparities in our model, as financially disadvantaged communities are likely to be more vulnerable during extreme weather events due to limited resources for cooling and emergency preparedness. [27]

Education also plays a key role in these correlations, with the "No Qualifications" variable showing a moderate positive correlation with unemployment (+0.38) and a weaker correlation with universal credit claims (+0.27), suggesting that lower educational attainment contributes to economic vulnerability. Additionally, "No Qualifications" has a moderate negative correlation with income (-0.22), further supporting the link between education and financial stability. Given that access to information and preparedness strategies is often linked to educational background, these findings suggest that areas with lower education levels may require additional outreach and support to effectively mitigate heatwave-related risks.[10]

Overall, this heatmap provides a view of how different socioeconomic and environmental factors interact and influence vulnerability. The strong correlations between overcrowding, economic disadvantage, and poorer environmental conditions underline key areas for intervention. Policies aimed at reducing unemployment, improving housing conditions, and addressing environmental concerns could significantly enhance resilience against heatwaves and power outages. Additionally, targeted support for both younger and older populations, particularly in terms of access to cooling resources and emergency response systems, could help bridge existing inequalities. [31]

3.7 Evaluation

3.7.1 Strengths

1. Scalability for Large Datasets:

As the dataset for the Population Demographics, Primary Method Travel to Work, and Total Number of Dwellings by type had a fairly large dataset, other algorithms such as clustering algorithms may not work as efficiently. PCA is much more scalable for larger datasets because its efficiency lies in reducing the dimensions of the data to capture the reduced set of uncorrelated principal components with the most variance, allowing a significant reduction in the complexity of the data. As a result, this allows the data to be easier to manage and interpret, hence improving the efficiency when creating a formula for the Vulnerability Score.

2. Combines Variance of Correlated Features:

In the dataset that we were given, the unemployment rate of residents aged 16 and over and percentage of Universal Credit Claimants may be highly correlated. The first principal component would likely capture the combined variance of both the unemployment rate and the claimant rate when the set of principal components are created. As a result, there would be a reduction in redundancy as well as a more efficient and simplified representation of data. On top of that, it would also mitigate the issue of multicollinearity, which could cause instability in our formula because a small fluctuation in data would lead to a large change in the resulting output of the formula.

3.7.2 Weaknesses

1. May Lead to Data Loss:

Whilst PCA may be efficient in capturing the components with high variance such as age, it would inevitably lead to lower variance data being overlooked. For instance, Housing Types and Quality, as older housing may be more vulnerable to heatwaves due to increased heat retention. As a result, our vulnerability score could be more skewed because it does not take lower variance components into account, even if it may be a significant factor in determining the final output. This, in turn, could make decisions made about allocating resources misjudged as they would not fully account for all relevant vulnerability factors. Despite this, one solution that we could incorporate would be to retain more components.

2. Sensitive to Scaling:

If features have vastly varying ranges, for example, the total population in a constituency may be measured in hundreds of thousands, whilst the Average Environmental Score is measured in decimals

below 1, PCA may offer components with a larger scale more prominent significance than small scale components without regard to their importance. This would lead to bias in our formula, which would lead to unreliable results. Therefore, if the data is not standardized, biased results may occur.

3. Assumes Normal Distribution of Data:

If extreme outliers are present, the variance of the data would be assumed to be constant. As a result, unreliable results may be presented because the true distribution of the data would not be represented effectively. Therefore, there is a need to clean the data before processing it.

References and appendices follow.

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5 Code

5.1 Question 1 Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import cross_val_score
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split #Importing all modules

df = pd.read_csv("dwellingsbirminghamcleaned.csv")

features = ["Stories", "Units in structure", "Size of unit (sqft)", "Year Structure Built",
            "Rooms", "People", "Shade"]

corr = df.corr() #Correlation heatmap for dwelling variables

plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap for Dwelling Variables")

plt.savefig("Correlation Heatmap for Dwelling Variables.jpg", dpi=300)

plt.show() #Plotting it using matplotlib

means = df[features].mean() #Working out means and standard devs to be used in standardisation
std_devs = df[features].std()

scaler = StandardScaler()
X_scaled = scaler.fit_transform(df) #Scaling using normal distribution

pca = PCA(n_components=2)
pca.fit(X_scaled)
loadings = pd.DataFrame(pca.components_.T,
                        index=df.columns,
                        columns=["PC1", "PC2"]) #Eigenvalues computation for each variable

print("Explained Variance Ratio:", pca.explained_variance_ratio_)

print("\nPCA Loadings:")
print(loadings)

#Weights for PC1 and PC2 based on explained variance ratios
pc1weight = pca.explained_variance_ratio_[0]
pc2weight = pca.explained_variance_ratio_[1]

#Coefficient calculations using eigenvalues and explained variance ratios
stories_coeff = -0.134179 * pc1weight + 0.268377 * pc2weight
units_coeff = 0.435329 * pc1weight + 0.114917 * pc2weight
size_coeff = -0.470718 * pc1weight + 0.135689 * pc2weight
year_coeff = 0.452023 * pc1weight + 0.204705 * pc2weight
rooms_coeff = -0.302813 * pc1weight + 0.605796 * pc2weight
people_coeff = 0.208838 * pc1weight + 0.698108 * pc2weight
shade_coeff = 0.480911 * pc1weight + -0.010448 * pc2weight

print("Stories coeff", stories_coeff, "Units coeff",
      units_coeff, "Size coeff", size_coeff, "Year coeff", year_coeff,
      "Rooms coeff", rooms_coeff, "People coeff", people_coeff, "Shade coeff", shade_coeff)

def outdoor_to_indoor(T_outdoor, stories, units, size, year, rooms, people, shade):
    # Standardisation of all values so that they have equal
    # weighting before applying the coefficients
    Stories_std = (stories - means["Stories"]) / std_devs["Stories"]
    Units_std = (units - means["Units in structure"]) / std_devs["Units in structure"]
    Size_std = (size - means["Size of unit (sqft)"]) / std_devs["Size of unit (sqft)"]
    Year_std = (year - means["Year Structure Built"]) / std_devs["Year Structure Built"]
```

```

Rooms_std = (rooms - means["Rooms"]) / std_devs["Rooms"]
People_std = (people - means["People"]) / std_devs["People"]
Shade_std = shade
sum_of_weights = Stories_std * stories_coeff +
Units_std * units_coeff + Size_std * size_coeff + Year_std
* year_coeff + Rooms_std * rooms_coeff + People_std
* people_coeff + Shade_std * shade_coeff
k = (1/(1 + np.e ** ((Stories_std * stories_coeff) * -1)))
+ (1/(1 + np.e ** ((Size_std * size_coeff) * -1)))
+ (1/(1 + np.e ** ((Year_std * year_coeff) * -1)))
+ (1/(1 + np.e ** ((Rooms_std * rooms_coeff) * -1)))
+ (1/(1 + np.e ** ((People_std * people_coeff) * -1)))
+ (1/(1 + np.e ** ((Shade_std * shade_coeff) * -1)))
k = k/7      #Calculation of thermal constant for the house (k)
t_in = T_outdoor + (T_outdoor - 24) * ((np.e) ** (k * -1))
return t_in

file_name = "Heatwaves"
encoding = "UTF-8"

file_path = f"{file_name}.csv"
df = pd.read_csv(file_path, encoding=encoding)

corr = df.corr()

plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap for Weather Variables")
plt.savefig("Correlation Heatmap.jpg", dpi=300)
plt.show()

hourly_datasets = {}
for hour in range(24):
    hourly_datasets[hour] = df[df['Time'] == hour]

hourly_means = df.groupby('Time')[['Dew Point (°C)', "Humidity (%)",
    'Wind Speed (miles per hour)']].mean() #Splits dataset by hour and calculates mean for each hour

def create_model(hour):
    df = hourly_datasets[hour]

    y = df["Temperature (°C)"]
    x = df[["Dew Point (°C)", "Humidity (%)", "Wind Speed (miles per hour)"]]
    x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
    scaler = StandardScaler()
    x_train_scaled = scaler.fit_transform(x_train)
    x_test_scaled = scaler.transform(x_test)

    model = RandomForestRegressor(n_estimators=100, random_state=42)
    #Creates Random Forest regressor to predict temperature
    model.fit(x_train_scaled, y_train)

    y_pred = model.predict(x_test_scaled)
    r2 = r2_score(y_test, y_pred)
    #print(f"R-squared value: {r2}")

    mse = mean_squared_error(y_test, y_pred)
    #print(f"MSE: {mse}")

    return model

predictions = []
for i in range(24):
    model = create_model(i)
    hour_data = hourly_means.loc[i]
    input_features = [[hour_data['Dew Point (°C)'], hour_data['Humidity (%)'],
        hour_data['Wind Speed (miles per hour)']]
    input_features = [[hour_data['Dew Point (°C)'], hour_data['Humidity (%)'],
        hour_data['Wind Speed (miles per hour)']]
    prediction = model.predict(input_features)[0]
    print(i, prediction)
    predictions.append(prediction) #Working out outdoor temperature predictions

```

```

homes_data = {"Home 1": {"Stories": 2, "Units in structure": 2,
    "Size of unit (sqft)": 1033, "Year Structure Built": 1940,
    "Rooms": 2, "People": 2, "Shade": 1},
    "Home 2": {"Stories": 3, "Units in structure": 9,
    "Size of unit (sqft)": 630, "Year Structure Built": 1980,
    "Rooms": 2, "People": 3, "Shade": 1},
    "Home 3": {"Stories": 2, "Units in structure": 6,
    "Size of unit (sqft)": 800, "Year Structure Built": 2010,
    "Rooms": 3, "People": 4, "Shade": 1},
    "Home 4": {"Stories": 3, "Units in structure": 1,
    "Size of unit (sqft)": 1500, "Year Structure Built": 1939,
    "Rooms": 4, "People": 3, "Shade": 0}}

house_name = {1: "Semi-detached whole house or bungalow", 2: "Flat (3rd floor)",
    3: "Maisonette", 4: "Terraced whole house or bungalow"}

for home in range(1, 5):
    indoor_predictions = []
    for i in range(24):
        indoor_predictions.append(outdoor_to_indoor(predictions[i],
            homes_data[f"Home {home}"]["Stories"],
            homes_data[f"Home {home}"]["Units in structure"],
            homes_data[f"Home {home}"]["Size of unit (sqft)"],
            homes_data[f"Home {home}"]["Year Structure Built"],
            homes_data[f"Home {home}"]["Rooms"], homes_data[f"Home {home}"]["People"],
            homes_data[f"Home {home}"]["Shade"]))
    indoor_predictions = [i + 2.5 for i in indoor_predictions]

    hours = list(range(24))
    print(indoor_predictions)
    plt.figure(figsize=(10, 5))
    plt.plot(hours, predictions, marker='o', linestyle='--', color='black',
        label="Outdoor Temperature", linewidth=2)
    plt.plot(hours, indoor_predictions, marker='o', linestyle='-', color='b',
        label="Indoor Temperature")

    plt.xlabel("Hour of the Day")
    plt.ylabel("Predicted Indoor Temperature (°C)")
    plt.title(f"Predicted Indoor Temperature vs. Hour of the Day for {house_name[home]}")

    plt.xticks(hours)
    plt.grid(True, linestyle="--", alpha=0.6)
    plt.legend()

    plt.savefig(f"Indoor Temperature vs. Hour of the day for {house_name[home]}.jpg", dpi=300)

    plt.show()

homes_data = {
    "Semi-detached whole house or bungalow": {"Stories": 2, "Units in structure": 2,
        "Size of unit (sqft)": 1033, "Year Structure Built": 1940, "Rooms": 2,
        "People": 2, "Shade": 1},
    "Flat (on 3rd floor)": {"Stories": 3, "Units in structure": 9,
        "Size of unit (sqft)": 630, "Year Structure Built": 1980, "Rooms": 2,
        "People": 3, "Shade": 1},
    "Maisonette": {"Stories": 2, "Units in structure": 6,
        "Size of unit (sqft)": 800, "Year Structure Built": 2010, "Rooms": 3,
        "People": 4, "Shade": 1},
    "Terraced (including end-terrace) whole house or bungalow": {"Stories": 3,
        "Units in structure": 1, "Size of unit (sqft)": 1500,
        "Year Structure Built": 1939, "Rooms": 4, "People": 3, "Shade": 0},
}

hours = list(range(24))
plt.figure(figsize=(10, 5))

colors = ['b', 'g', 'r', 'c']

for idx, (home, attributes) in enumerate(homes_data.items(), start=1):
    indoor_predictions = [
        outdoor_to_indoor(

```

```

        predictions[i],
        attributes["Stories"],
        attributes["Units in structure"],
        attributes["Size of unit (sqft)"],
        attributes["Year Structure Built"],
        attributes["Rooms"],
        attributes["People"],
        attributes["Shade"],
    ) + 2.5
    for i in range(24)
]
plt.plot(hours, indoor_predictions, marker='o',
linestyle='-', color=colors[idx-1], label=f"{home}")

plt.xlabel("Hour of the Day")
plt.ylabel("Predicted Indoor Temperature (°C)")
plt.title("Predicted Indoor Temperature vs. Hour of the Day for Multiple Homes")
plt.xticks(hours)
plt.grid(True, linestyle="--", alpha=0.6)
plt.legend()
plt.savefig("Predicted Indoor Temperature vs. Hour of the Day for Multiple Homes.jpg", dpi=300)
plt.show()

```

5.2 Question 2 Code

```

"""Import Libraries"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import pmdarima as ar
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import MinMaxScaler

"""Get Data from CSV"""
file_name = "Birmingham - Electricity Consumption"
encoding = "UTF-8"

file_path = f"/Data/{file_name}.csv"
df = pd.read_csv(file_path, encoding=encoding,
parse_dates=True, index_col="Date").sort_index()
df.index = pd.to_datetime(df.index)

df = df.dropna()

"""Get additional data from CSV"""
file_path = "/content/drive/My Drive/M3 Challenge/Data/Q2 Additional Data/temp.csv"
average_temp_df = pd.read_csv(file_path, encoding=encoding).sort_index()

file_path = "/content/drive/My Drive/M3 Challenge/Data/Q2 Additional Data/prices.csv"
electricity_prices_df = pd.read_csv(file_path, encoding=encoding).sort_index()

file_path = "/content/drive/My Drive/M3 Challenge/Data/Q2 Additional Data/hot days.csv"
no_hot_days_df = pd.read_csv(file_path, encoding=encoding).sort_index()

file_path = "/content/drive/My Drive/M3 Challenge/Data/Q2 Additional Data/population.csv"
population_df = pd.read_csv(file_path, encoding=encoding).sort_index()

"""Create Models for Additional Data"""
x = average_temp_df["Year"].values.reshape(-1, 1)
y = average_temp_df['Average Temperature'].values

average_temp_model = LinearRegression()
average_temp_model.fit(x, y)
predict_temp = lambda year: average_temp_model.predict(np.array([[year]]))

```

```

x = electricity_prices_df["Year"].values.reshape(-1, 1)
y = electricity_prices_df['Credit: Unit cost (Pence per kWh)'].values

electricity_prices_model = LinearRegression()
electricity_prices_model.fit(x, y)
predict_prices = lambda year: electricity_prices_model.predict(np.array([[year]]))

x = population_df["Year"].values.reshape(-1, 1)
y = population_df['Population'].values

population_model = LinearRegression()
population_model.fit(x, y)
predict_population = lambda year: population_model.predict(np.array([[year]]))

hot_days_model = ARIMA(no_hot_days_df['Number of Hot Days'], order=(1, 1, 1))
hot_days_model = hot_days_model.fit()

def predict_hot_days(year):
    year = round(year, 3)
    if int(year) in no_hot_days_df['Year'].values
    and int(year + 1) in no_hot_days_df['Year'].values:
        before = no_hot_days_df.loc[no_hot_days_df['Year'] == int(year)
        , 'Number of Hot Days'].iloc[0]
        after = no_hot_days_df.loc[no_hot_days_df['Year'] == int(year+1)
        , 'Number of Hot Days'].iloc[0]
        change = year - int(year)
        return before + ((after - before) * change)
    steps_ = int(year) - int(no_hot_days_df["Year"].iloc[-1])
    if steps_ > 0:
        forecast_values = hot_days_model.forecast(steps=steps_)
        return forecast_values.iloc[-1]
    else:
        return no_hot_days_df['Number of Hot Days'].iloc[-1]

"""Visualise Time Series & Check for Stationarity"""
time_series = df["Electricity Consumption (kWh)"].iloc[:-20]
original_time_series = time_series.copy()
time_series = time_series.diff().dropna()
result = adfuller(time_series)
print("ADF Statistic:", result[0])
print("p-value:", result[1])
original_time_series.plot()
plt.show()

"""Plot ACF and PACF to determine AR and MA Orders (p, d, q)"""
plot_pacf(time_series, lags=20)
plot_acf(time_series, lags=20)
plt.show()

"""Find Orders"""
train_exog = pd.DataFrame({
    'temp': [predict_temp(item.year
        + ((item.month-1)/12))[0] for item in time_series.index],
    'prices': [predict_prices(item.year
        + ((item.month-1)/12))[0] for item in time_series.index],
    'population': [predict_population(item.year
        + ((item.month-1)/12))[0] for item in time_series.index],
    'hot_days': [predict_hot_days(item.year
        + ((item.month-1)/12)) for item in time_series.index]
})
weighted_time_series = time_series.copy()
covid_years = weighted_time_series.index[(weighted_time_series.index.year >= 2020)
    & (weighted_time_series.index.year <= 2022)]
weighted_time_series[covid_years] *= 0.5
amd = ar.auto_arima(
    weighted_time_series, exogenous=train_exog, trace=True, suppress_warnings=True
)
print(amd.summary())

"""Create ARIMA Model"""
order = amd.order

```

```

train_exog.index = time_series.index
train = time_series
test = time_series.iloc[-30:]
weights = np.ones(len(train))
covid_years = train.index[(train.index.year >= 2020) & (train.index.year <= 2022)]
weights[train.index.isin(covid_years)] = 0
model = SARIMAX(train, exog=train_exog, order=order, seasonal_order=(*order, 12))
model = model.fit(weights=weights)
print(model.summary())

"""Evaluate Model"""
predictions = model.predict(start=len(train), end=len(train)+len(test)-1, typ="levels")
r2 = r2_score(test, predictions)
print(f"r-squared value: {r2}")
mse = mean_squared_error(test, predictions)
print(f"MSE: {mse}")
predictions = model.predict(start=len(train), end=len(train)+len(test)-1, typ="levels")
r2 = r2_score(test, predictions)
print(f"r-squared value: {r2}")
mse = mean_squared_error(test, predictions)
print(f"MSE: {mse}")

"""Predict Data"""
steps = 300
forecast_dates = pd.date_range(time_series.index[-1], periods=steps + 1, freq="M")
forecast_exog = pd.DataFrame({
    'temp': [predict_temp(item.year
        + ((item.month)/12))[0] for item in forecast_dates],
    'prices': [predict_prices(item.year
        + ((item.month)/12))[0] for item in forecast_dates],
    'population': [predict_population(item.year
        + ((item.month)/12))[0] for item in forecast_dates],
    'hot_days': [predict_hot_days(item.year
        + ((item.month)/12)) for item in forecast_dates]
})
forecast_exog.index = forecast_dates
forecast = model.predict(start=len(time_series), end=len(time_series)+steps,
    exog=forecast_exog, typ="levels")
forecast = forecast.cumsum()
forecast_dates = pd.date_range(time_series.index[-1]-pd.DateOffset(months=1),
    periods=len(forecast)+1, freq="M")[1:]
forecast_dates = pd.to_datetime(forecast_dates)

"""Plot Graph"""
x_label = "Year"
y_label = "Energy Consumption / GWh"
title = "Projected Energy Consumption in Birmingham (monthly)"
time_series.index = pd.to_datetime(time_series.index)
p_time_series = original_time_series / 1000000
p_forecast = forecast / 1000000
plt.figure(figsize=(8, 6))
plt.plot(p_time_series, linestyle="-", color="blue", label="Training")
plt.plot(forecast_dates, p_forecast, linestyle="-", color="red", label="Predicted")
plt.xlabel(x_label)
plt.ylabel(y_label)
plt.title(title)
plt.legend()
plt.show()

```

5.3 Question 3 Code

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import cross_val_score
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

```

```

import warnings
warnings.filterwarnings("ignore") #Importing all modules

df = pd.read_csv("/content/drive/My Drive/M3 Challenge/Data/Vulnerability.csv")

print(df.head())

modified_df = df.drop(columns=["Constituency"])
#Dropping constituency from data frame since not used in formula
scaler = StandardScaler()
X_scaled = scaler.fit_transform(modified_df)

pca = PCA(n_components=2)
pca.fit(X_scaled)
loadings = pd.DataFrame(pca.components_.T,
index=modified_df.columns, columns=["PC1", "PC2"])

print("Explained Variance Ratio:", pca.explained_variance_ratio_)

print("\nPCA Loadings:")
print(loadings)

pc1weight = pca.explained_variance_ratio_[0]
pc2weight = pca.explained_variance_ratio_[1]

population_coeff = pc1weight * 0.267104 + pc2weight * 0.281939
population_under_15_coeff = pc1weight * 0.360466 + pc2weight * 0.161359
population_65over_coeff = pc1weight * -0.336981 + pc2weight * 0.348304
income_coeff = pc1weight * -0.316867 + pc2weight * 0.079504
overcrowded_coeff = pc1weight * 0.318233 + pc2weight * -0.385954
unemployment_coeff = pc1weight * 0.211376 + pc2weight * 0.646686
universal_credit_coeff = pc1weight * -0.344344 + pc2weight * 0.163373
no_qualifications_coeff = pc1weight * 0.171804 + pc2weight * 0.374619
transport_coeff = pc1weight * -0.379173 + pc2weight * 0.144125
environment_coeff = pc1weight * 0.383335 + pc2weight * 0.108954

numeric_cols = df.select_dtypes(include=np.number).columns
scaler = MinMaxScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
#Scaler applied to all data
vulnerability_scores = []

for index, row in df.iterrows():
    constituency = row["Constituency"]

    print(constituency)

    vulnerability_score = row["Population"] * population_coeff + row["U15"] * population_under_15_coeff +
    * population_65over_coeff + row["Income"] * income_coeff + row["Overcrowding"]
    * overcrowded_coeff + row["Unemployment"] * unemployment_coeff + row["Universal Credit"]
    * universal_credit_coeff + row["No Qualifications"] * no_qualifications_coeff
    + row["\nPercentage of households with 2+ cars or vans"]
    * transport_coeff + row["Environmental Score"] * environment_coeff
    vulnerability_scores.append(vulnerability_score) #Sum of weighted values

scaler = MinMaxScaler()
vulnerability_scores = scaler.fit_transform(np.array(vulnerability_scores).reshape(-1, 1))

print(vulnerability_scores)

import matplotlib.pyplot as plt

data = pd.DataFrame({
    "Constituency": [
        "Birmingham Edgbaston", "Birmingham Erdington"
        , "Birmingham Hall Green and Moseley",
        "Birmingham Hodge Hill and Solihull North"
        , "Birmingham Ladywood", "Birmingham Northfield",
        "Birmingham Perry Barr", "Birmingham Selly Oak"
        , "Birmingham Yardley", "Sutton Coldfield"
    ],

```

```

        "Vulnerability Score": [0.3427, 1, 0.5958, 0.7875,
                                0.7944, 0.423, 0.8918, 0.5171, 0.8786, 0]
    })
    data = data.sort_values(by="Vulnerability Score", ascending=True)

    plt.figure(figsize=(10, 6))
    plt.barh(data["Constituency"], data["Vulnerability Score"], color="red")
    #Setting bar parameters

    plt.xlabel("Vulnerability Score")
    plt.ylabel("Constituency")
    plt.title("Vulnerability Scores of Birmingham Constituencies")

    #Show plot

    plt.savefig("Q3 Bar Chart.jpg", dpi=300)

    plt.show()

    """Create Choropleth Map"""

    import geopandas as gpd
    import pandas as pd
    import folium
    import matplotlib.pyplot as plt

    shapefile_path = "/content/drive/My Drive/M3 Challenge/Data/map.geojson"
    birmingham_map = gpd.read_file(shapefile_path)

    data = pd.DataFrame({
        "Constituency": [
            "Birmingham Edgbaston", "Birmingham Erdington"
            , "Birmingham Hall Green and Moseley",
            "Birmingham Hodge Hill and Solihull North", "Birmingham Ladywood"#
            , "Birmingham Northfield",
            "Birmingham Perry Barr", "Birmingham Selly Oak"
            , "Birmingham Yardley", "Sutton Coldfield"
        ],
        "Vulnerability Score": [0.3427, 1, 0.5958, 0.7875, 0.7944, 0.423, 0.8918, 0.5171, 0.8786, 0]
    })

    birmingham_map = birmingham_map.merge(data, left_on="pcon24nm", right_on="Constituency")

    fig, ax = plt.subplots(1, 1, figsize=(10, 6))
    birmingham_map.plot(column="Vulnerability Score", cmap="Reds",
                        linewidth=0.8, edgecolor="black", legend=True, ax=ax)

    plt.title("Vulnerability Score across Birmingham Constituencies")
    plt.axis("off") # Hide axes

    plt.savefig("Q3 Choropleth Map.jpg", dpi=300)

    plt.show()

    """Creating a Correlation Heatmap"""

    correlation_matrix = df[numeric_cols].corr()

    plt.figure(figsize=(12, 10))
    sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=.5)
    plt.title("Correlation Heatmap of Factors")

    plt.savefig("Q3 Correlation Heatmap of Factors.jpg", dpi=300)

    plt.show()

```